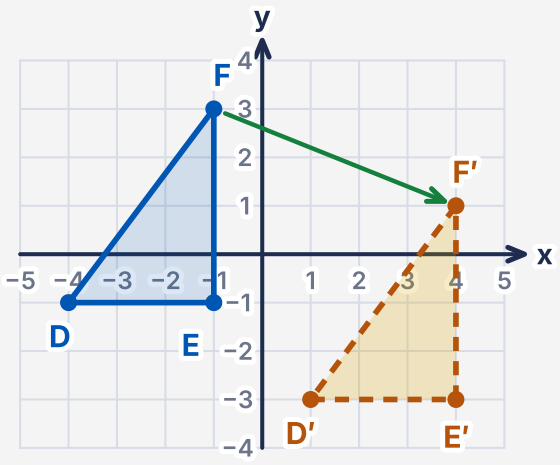
📝 Unit 8 Practice 2 — Transformations

Complete independently · NC.8.G.2–4 · Show all coordinate work

ANSWER KEY

`Translation: (x, y) → (x + h, y + k) — slides without rotating or flipping`

Question 1 — Translate a Triangle

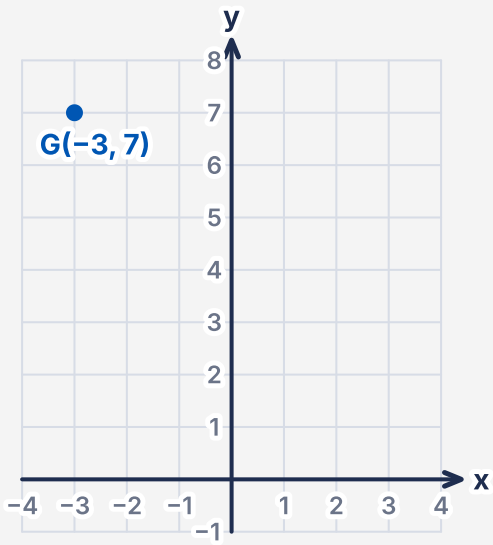


blue = pre-image · **gold dashed** = image

Triangle DEF was translated to D′E′F′. If D = (−4, −1) and D′ = (1, −3), identify the translation rule.

- ○ (x − 5, y + 2)
- ● **(x + 5, y − 2)**
- ○ (x + 2, y − 5)
- ○ (x + 5, y + 2)

Question 2



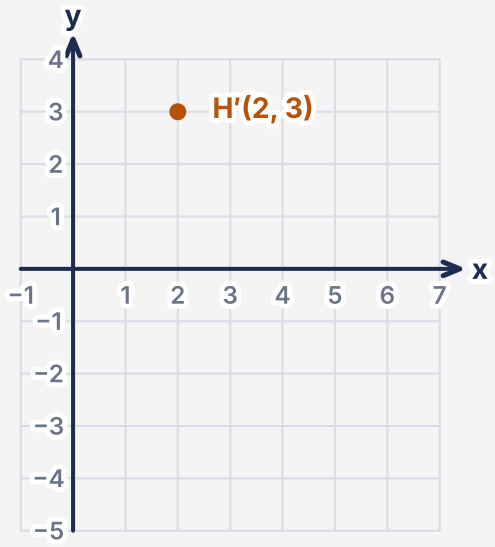
Rule: (x, y) → (x + 6, y − 5)

blue = pre-image · **gold dashed** = image

Translate G(−3, 7) using (x + 6, y − 5). Find G′.

- ● **G′(3, 2)**
- ○ G′(3, 12)
- ○ G′(−9, 12)
- ○ G′(3, −2)

Question 3



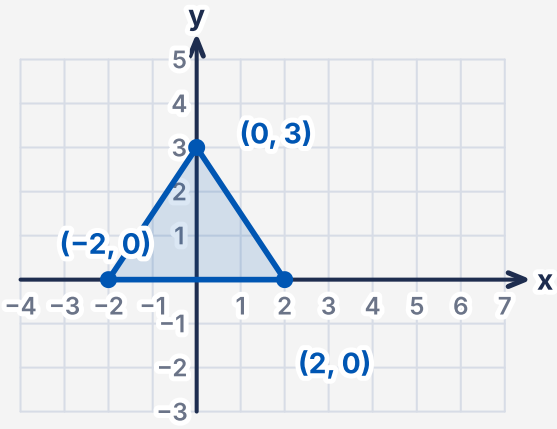
Rule: (x, y) → (x − 4, y + 7)

blue = pre-image · **gold dashed** = image

After rule (x − 4, y + 7), point H′ is at (2, 3). Find the original H.

- ● **H(6, −4)**
- ○ H(−2, 10)
- ○ H(−2, −4)
- ○ H(6, 10)

Question 4



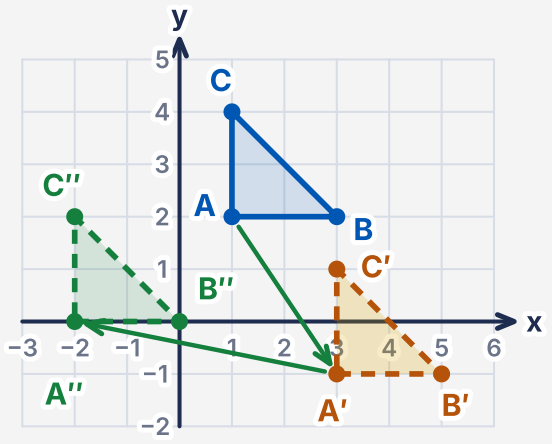
Rule: (x, y) → (x + 3, y − 1)

blue = pre-image · **gold dashed** = image

Triangle with vertices (−2,0),(2,0),(0,3) is translated (x+3, y−1). What is the new position of (0,3)?

- ● **(3,2)**
- ○ (−2,2)
- ○ (3,0)
- ○ (0,2)

Question 5



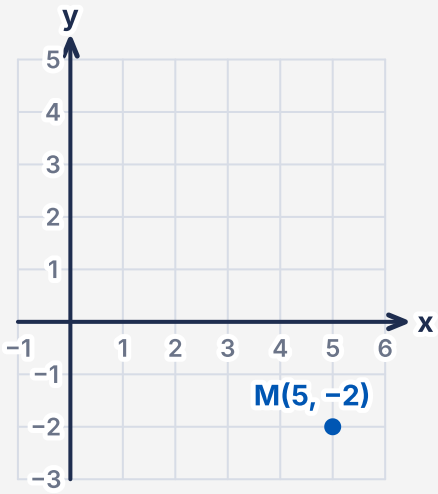
Step 1: (x, y) → (x + 2, y − 3)
Step 2: (x, y) → (x − 5, y + 1)

blue → **gold** (step 1) → **green** (step 2)

A figure is translated (x+2, y−3), then translated again (x−5, y+1). What single rule equals both translations combined?

- ● **(x−3, y−2)**
- ○ (x+3, y−2)
- ○ (x−7, y+4)
- ○ (x−3, y+2)

Question 6



Rule: slide 3 left, 6 up

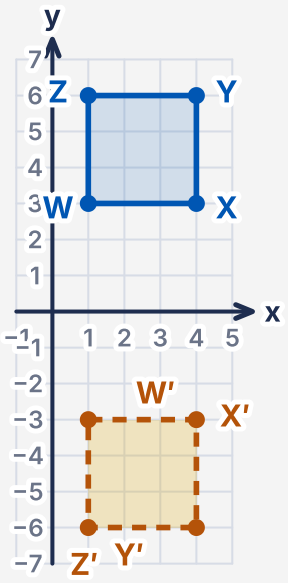
blue = pre-image · **gold dashed** = image

Point M(5, −2) is translated 3 units left and 6 units up. What are M′'s coordinates?

- ● **M′(2, 4)**
- ○ M′(8, 4)
- ○ M′(2, −8)
- ○ M′(8, −8)

```
x-axis reflection: (x,y)→(x,−y) | y-axis reflection: (x,y)→(−x,y)
```

Question 1 — Reflect a Rectangle

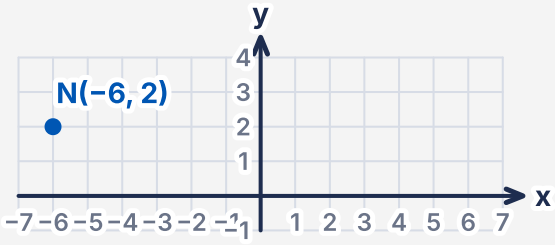


blue = pre-image · **gold dashed** = image

Rectangle WXYZ was reflected. What line of reflection was used?

- ○ y-axis
- ● **x-axis**
- ○ y = x
- ○ y = −1

Question 2



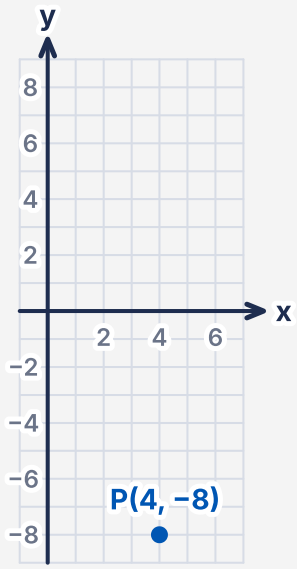
Rule: (x, y) → (−x, y)

blue = pre-image · **gold dashed** = image

Reflect N(−6, 2) over the y-axis.

- ● **N′(6, 2)**
- ○ N′(−6, −2)
- ○ N′(6, −2)
- ○ N′(2, −6)

Question 3



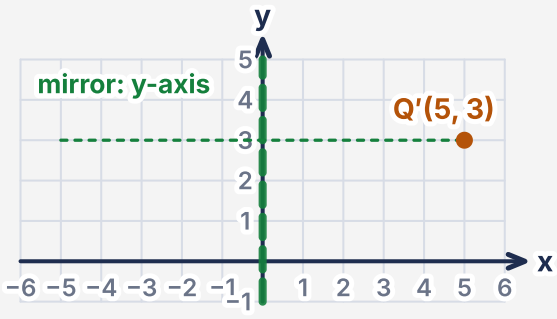
Rule: (x, y) → (x, −y)

blue = pre-image · **gold dashed** = image

Reflect P(4, −8) over the x-axis.

- ○ P′(−4, 8)
- ● **P′(4, 8)**
- ○ P′(−4, −8)
- ○ P′(8, 4)

Question 4



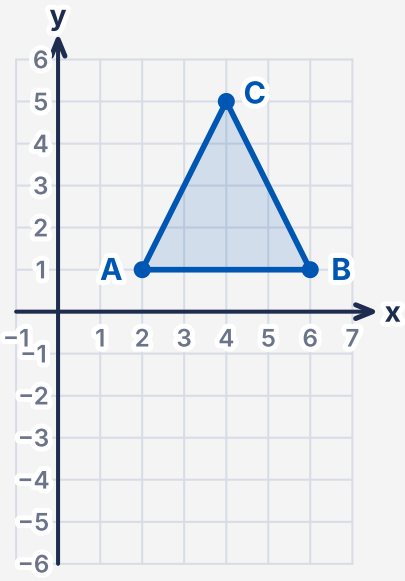
Rule: (x, y) → (−x, y)

blue = pre-image · **gold dashed** = image

A point Q′ after reflection over the y-axis is at (5, 3). Find original Q.

- ○ Q(5, 3)
- ● **Q(−5, 3)**
- ○ Q(5, −3)
- ○ Q(3, 5)

Question 5



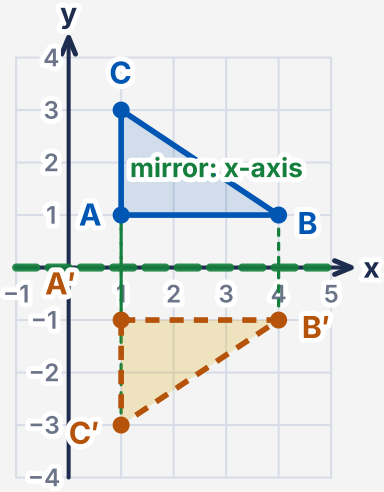
Rule: (x, y) → (x, −y)

blue = pre-image · **gold dashed** = image

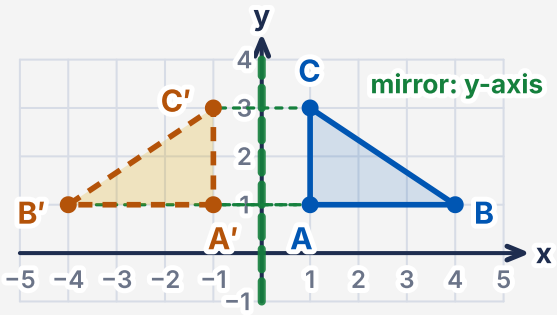
Triangle A(2,1), B(6,1), C(4,5) reflected over x-axis. Find B′.

- ○ B′(−6, 1)
- ● **B′(6, −1)**
- ○ B′(−6, −1)
- ○ B′(1, 6)

Question 6



over the x-axis



over the y-axis

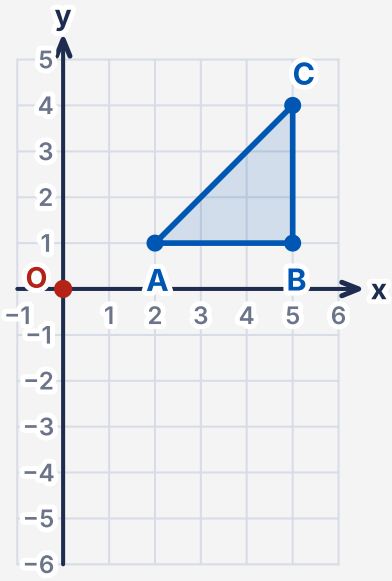
blue = pre-image · **gold dashed** = image

What do reflections over the x-axis and y-axis have in common?

- ○ Both flip x-coordinates
- ● **Both produce congruent (same size and shape) figures**
- ○ Both always move figures to a new quadrant
- ○ Both are non-rigid transformations

`90°CCW:(x,y)→(−y,x) | 180°:(x,y)→(−x,−y) | 270°CCW:(x,y)→(y,−x) | 90°CW=270°CCW`

Question 1 — 270° CCW Rotation



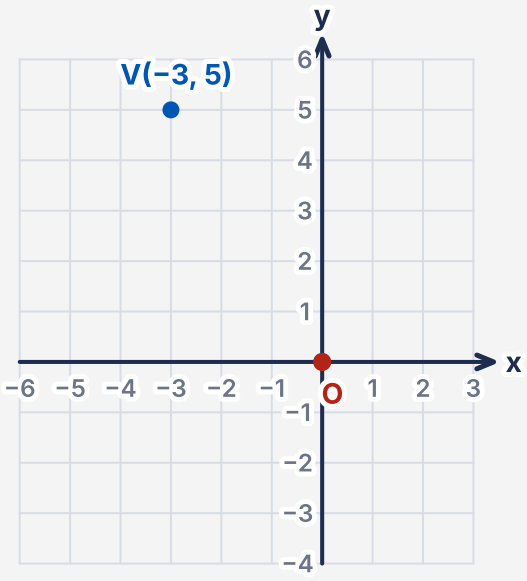
Rule: (x, y) → (y, −x)

blue = pre-image · **gold dashed** = image

Triangle ABC was rotated 270° CCW. If A = (2, 1), find A′.

- ○ A′(−1, 2)
- ● **A′(1, −2)**
- ○ A′(−2, −1)
- ○ A′(2, −1)

Question 2



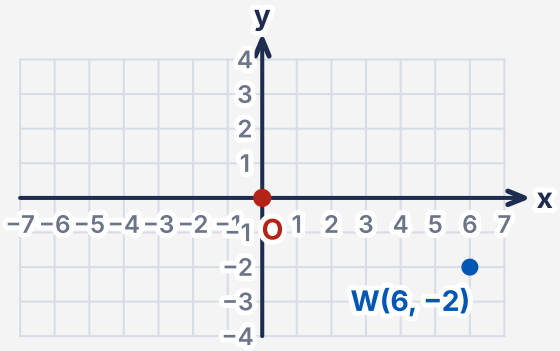
Rule: (x, y) → (−y, x)

blue = pre-image · **gold dashed** = image

Rotate V(−3, 5) by 90° CCW.

- ● **V′(−5, −3)**
- ○ V′(5, 3)
- ○ V′(3, 5)
- ○ V′(−5, 3)

Question 3



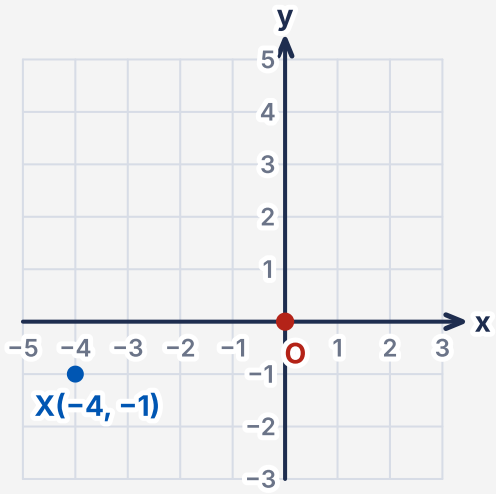
Rule: (x, y) → (−x, −y)

blue = pre-image · **gold dashed** = image

Rotate W(6, −2) by 180°.

- ● **W′(−6, 2)**
- ○ W′(2, 6)
- ○ W′(6, 2)
- ○ W′(−2, −6)

Question 4



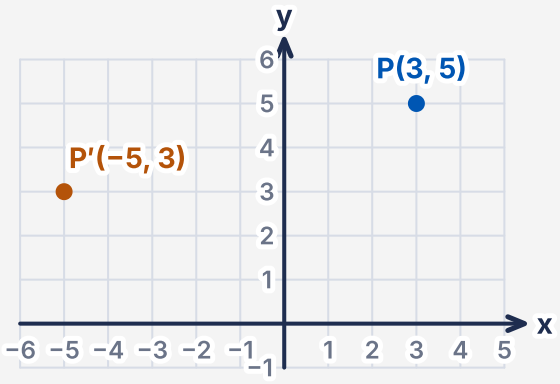
Rule: (x, y) → (y, −x)

blue = pre-image · **gold dashed** = image

Rotate X(−4, −1) by 90° CW (= 270° CCW).

- ○ X′(1, 4)
- ○ X′(4, 1)
- ● **X′(−1, 4)**
- ○ X′(1, −4)

Question 5

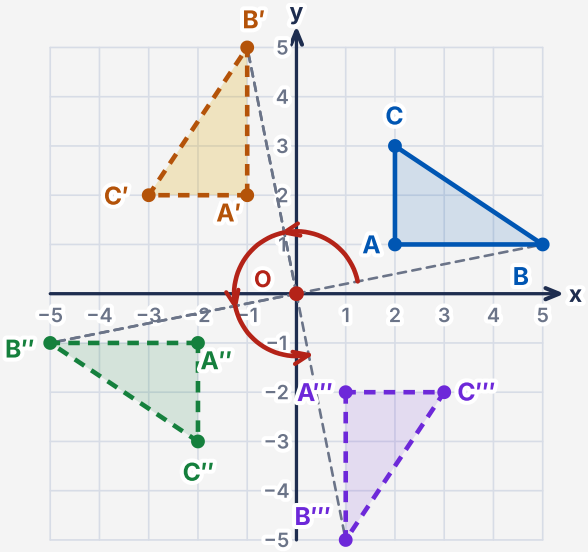


blue = pre-image · **gold dashed** = image

Which rotation maps (3, 5) to (−5, 3)?

- ○ 90° CW
- ○ 180°
- ● **90° CCW**
- ○ 270° CCW

Question 6



Step 1: (x, y) → (−y, x)
Step 2: (x, y) → (−y, x)
Step 3: (x, y) → (−y, x)
↻ each red arrow = one 90° CCW turn

blue → **gold** (step 1) → **green** (step 2) → **purple** (step 3)

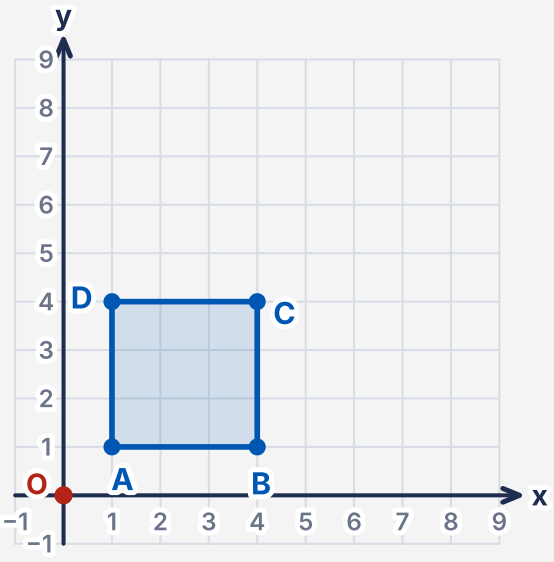
Each red arrow is one 90° CCW turn about the origin O.

A figure is rotated 90°CCW three times. What single rotation gives the same result?

- ● **270° CCW**
- ○ 90° CCW
- ○ 180°
- ○ 360°

```
Dilation: (x,y)→(kx,ky) | k>1=enlargement | 0<k<1=reduction | Similar figures
```

Question 1 — Dilation k = 2



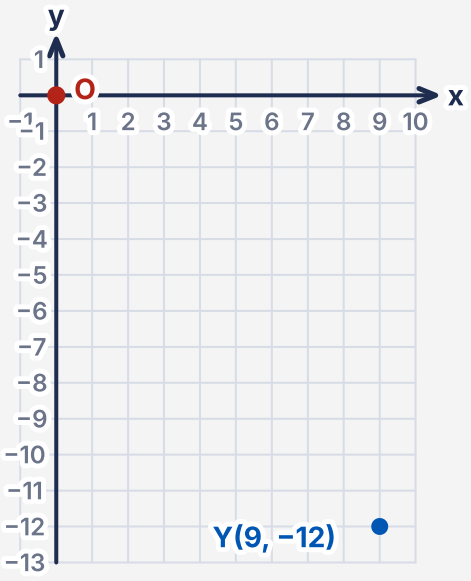
Rule: (x, y) → (2x, 2y)

blue = pre-image · **gold dashed** = image

Square ABCD was dilated with k = 2. If A = (1, 1), find A′.

- A′(1, 1)
- **A′(2, 2)**
- A′(3, 3)
- A′(0.5, 0.5)

Question 2



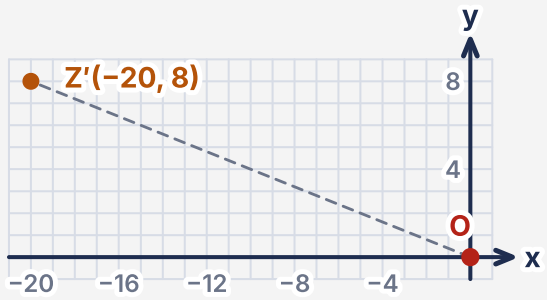
Rule: (x, y) → (⅓x, ⅓y)

blue = pre-image · **gold dashed** = image

Dilate Y(9, −12) with k = ⅓.

- **Y′(3, −4)**
- Y′(27, −36)
- Y′(3, 4)
- Y′(−3, 4)

Question 3



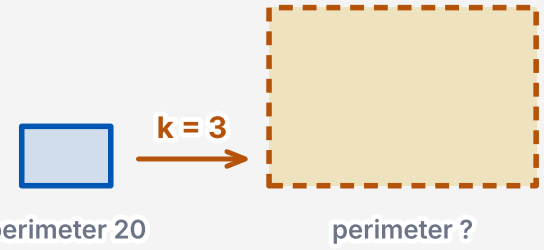
Rule: (x, y) → (4x, 4y)

blue = pre-image · **gold dashed** = image

The image Z′ is at (−20, 8). If k = 4, find original Z.

- **Z(−5, 2)**
- Z(−80, 32)
- Z(−5, −2)
- Z(5, 2)

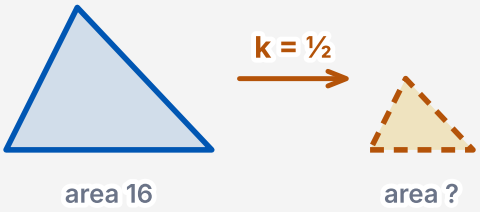
Question 4



Rectangle with perimeter 20 is dilated k = 3. New perimeter?

- **60**
- 23
- 180
- 9

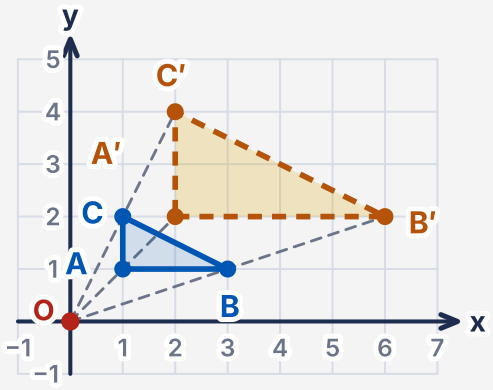
Question 5



Triangle area = 16 sq units. After dilation k = ½, what is the new area?

- 8 sq units
- **4 sq units**
- 32 sq units
- 64 sq units

Question 6



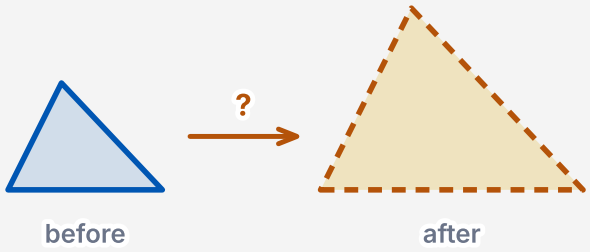
blue = pre-image · **gold dashed** = image

A dilation maps every point farther from the origin. This is an:

- **Enlargement (k > 1)**
- Reduction (0 < k < 1)
- Reflection
- Translation

Mixed Review: These questions combine all transformation types.

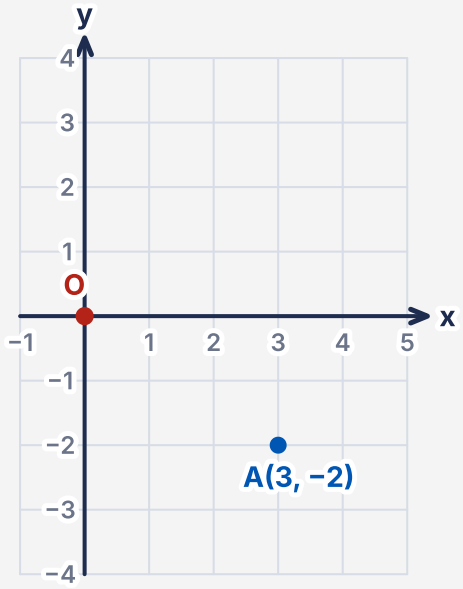
Question 1



Which transformation changes the SIZE of a figure?

- ○ Translation
- ○ Reflection
- ○ Rotation
- ● **Dilation**

Question 2



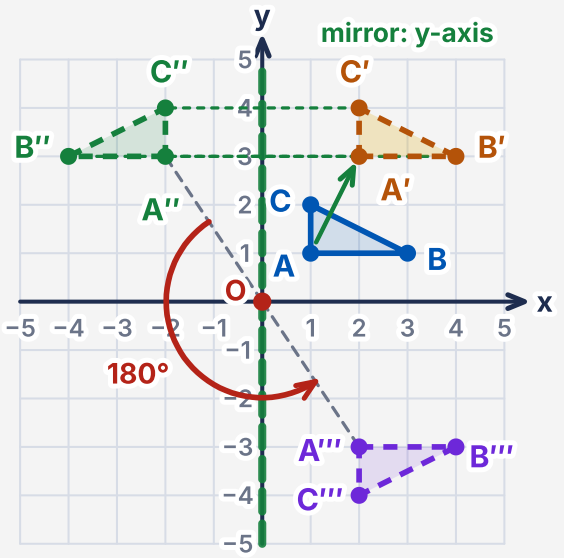
Step 1: (x, y) → (−y, x)
Step 2: (x, y) → (x, −y)

blue → **gold** (step 1) → **green** (step 2)

Point A(3, −2): apply rotation 90°CCW, then reflection over x-axis. Find A″.

- ○ (2, 3)
- ● **(2, −3)**
- ○ (−2, 3)
- ○ (−2, −3)

Question 3



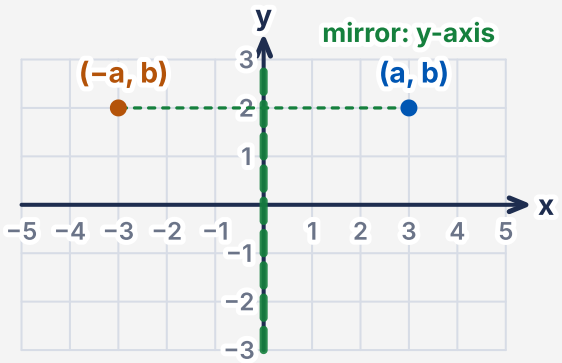
Step 1: slide 1 right, 2 up
Step 2: flip over y-axis
Step 3: turn 180°

blue → **gold** (step 1) → **green** (step 2) → **purple** (step 3)

Figure M is translated, reflected, then rotated. Is the result congruent or similar to M?

- ○ Similar — three transformations change size
- ● **Congruent — all three are rigid**
- ○ Similar — rotation changes shape
- ○ Cannot determine

Question 4



Step 1: (x, y) → (−x, y)
Step 2: (x, y) → (x, −y)

blue → **gold** (step 1) → **green** (step 2)

What is the result of reflecting (a, b) first over the y-axis, then over the x-axis?

- ○ (a, b)
- ○ (−a, b)
- ○ (a, −b)
- ● **(−a, −b)**

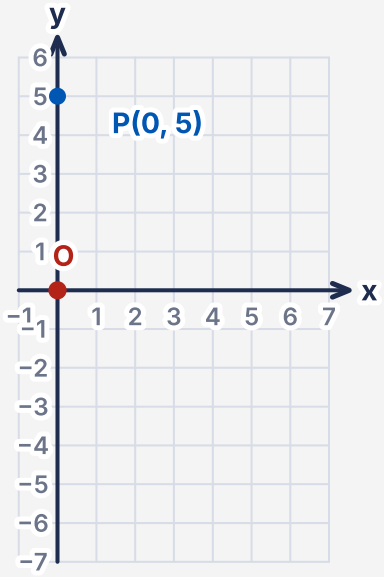
Question 5



A 10-unit segment is dilated k = 2.5. What is the new length?

- ● **25 units**
- ○ 12.5 units
- ○ 20 units
- ○ 4 units

Question 6



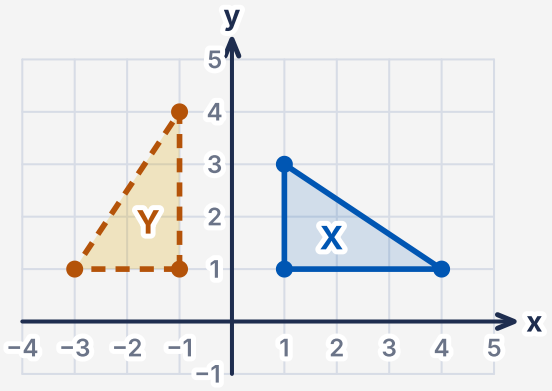
Step 1: (x, y) → (x + 3, y − 8)
Step 2: (x, y) → (2x, 2y)

blue → **gold** (step 1) → **green** (step 2)

Translate P(0, 5) by (x + 3, y − 8), then dilate k = 2. Find P″.

- ● **(6, −6)**
- ○ (3, −3)
- ○ (6, −16)
- ○ (3, −8)

Question 7

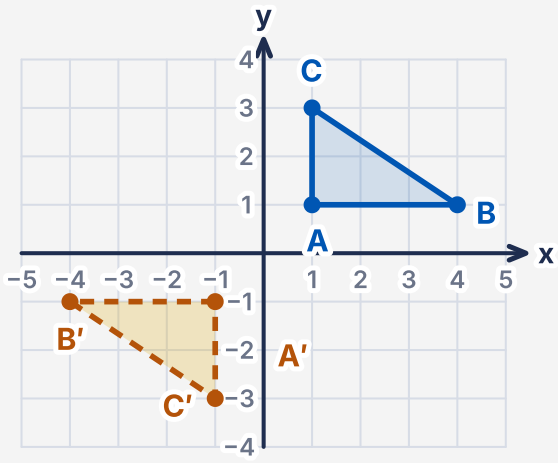


X and **Y** are congruent

Figures X and Y are congruent. Which transformation was NOT used?

- ○ Translation
- ● **Dilation**
- ○ Reflection
- ○ Rotation

Question 8



blue = pre-image · **gold dashed** = image

What transformation maps EVERY point (x, y) to (−x, −y)?

- ○ Reflection over x-axis
- ○ Reflection over y-axis
- ● **Rotation 180° about origin**
- ○ Translation

📝 Worked Examples

Study these step-by-step solutions before attempting the problems above.

Composite: translate △ABC — A(2,5), B(5,4), C(3,1) — by (x−3, y+1), then reflect over the x-axis

Step 1: Translate every vertex of △ABC by (x−3, y+1)

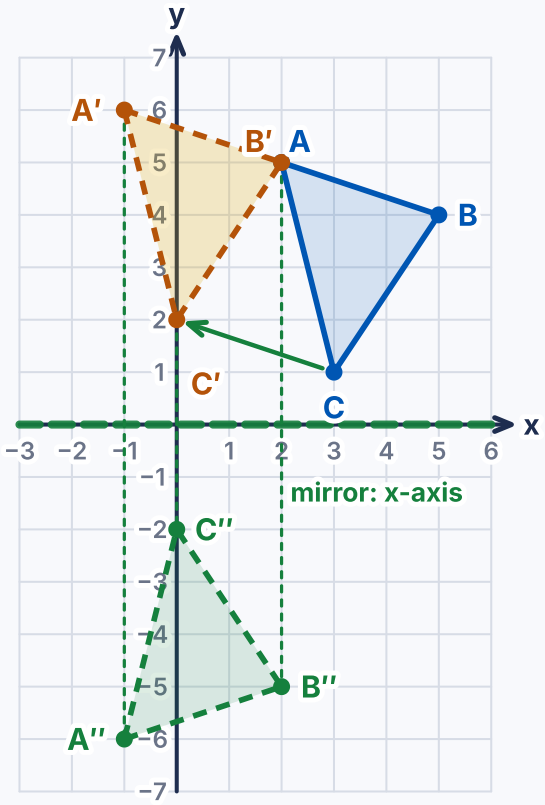
```
A(2,5)→A'(−1,6)    B(5,4)→B'(2,5)    C(3,1)→C'(0,2)
```

Step 2: Reflect every vertex over the x-axis: (x, −y)

```
A'(−1,6)→A"(−1,−6)    B'(2,5)→B"(2,−5)    C'(0,2)→C"(0,−2)
```

Step 3: Final image

```
△A"B"C": A"(−1,−6)    B"(2,−5)    C"(0,−2) ✓
```



Step 1: slide 3 left, 1 up
Step 2: flip over x-axis
A and B′ share the point (2, 5)

△ABC (solid blue) → translate → **△A′B′C′** (dashed gold) → reflect over x-axis → **△A″B″C″** (dashed green)

🔍 Common Mistakes

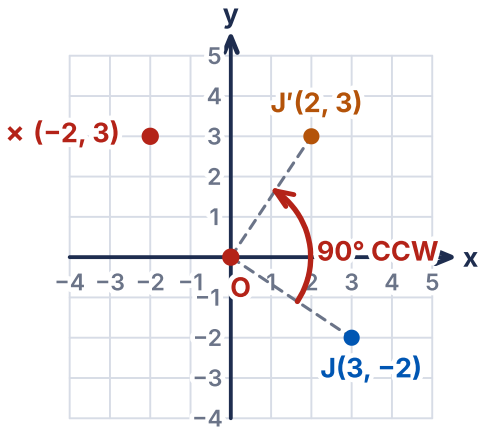
Can you spot what went wrong?

Rotate △JKL 90° CCW — vertex J is at (3,−2)

✗ **Student:** Student writes: (−2, 3)

✓ **Fix:** WRONG! Rule is (−y, x). y=−2, so −y = 2. Correct: (2, 3)

💡 *Be careful with negatives in rotation rules. −(−2) = +2.*



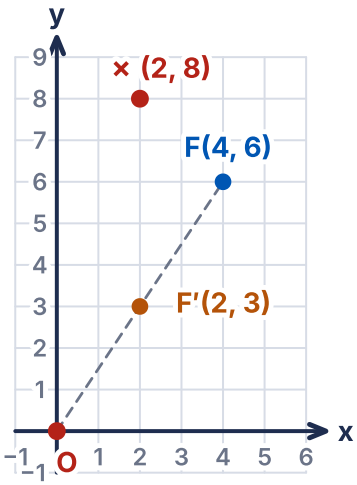
gold = correct · red × = student

Dilate △FGH by k=½ — vertex F is at (4,6)

✗ **Student:** Student writes: (2, 8)

✓ **Fix:** WRONG! ½ × 4 = 2 ✓ but ½ × 6 = 3, not 8. Correct: (2, 3)

💡 *Multiply BOTH coordinates by the scale factor.*



gold = correct · red × = student

⚡ Key Tips for This Unit

✅ Translation: (x+a, y+b) — slide. CONGRUENT.

✅ Reflection: x-axis → negate y. y-axis → negate x. CONGRUENT.

✅ Rotation 90° CCW: (−y, x). 180°: (−x,−y). 270° CCW: (y,−x). CONGRUENT.

✅ Dilation: (kx, ky). SIMILAR (not congruent unless k=1).